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Mock Paper 1

The R2Drew2 Edition

One of the eight most important mock papers in the world.

75 minutes • 20 questions

- You must answer all 20 questions. There is no penalty for guessing.
- Each question has between four and eight options, of which exactly one is correct.
- Calculators are **not** permitted.
- You do not need to show your working out.
- This paper contains no questions about the Large Data Set.

- 1 The sum of the first n terms of a sequence is given by the formula $S_n = 5n^2 + kn$, where k is a constant. The 6th term of the sequence is 52. Find the value of k .

A -13

B -8

C -3

D 3

E 22

2 Let $P(x)$ be defined by:

$$P(x) = x^{2027} - 4x^{2026} + 6$$

When $P(x)$ is divided by $x^2 - 1$, the remainder is $R(x) = ax + b$.
What is $R(x)$?

A $(x + 2)$

B $(x - 2)$

C $(-x + 2)$

D $(2x + 1)$

- 3** Find the value of the constant k such that the equation

$$(x^2 - 4x)^2 - 2k(x^2 - 4x) + k^2 - 9 = 0$$

has exactly three distinct real solutions for x .

A -7

B -5

C -1

D 3

E 5

F 7

- 4 A sequence u_n is defined by $u_0 = 3$ and the recurrence relation

$$u_{n+1} = 1 - \frac{1}{u_n}$$

for $n \geq 0$.

Find the value of the sum $S = \sum_{n=0}^{25} u_n$.

A $\frac{51}{2}$

B 26

C $\frac{85}{3}$

D $\frac{57}{2}$

E 29

F 37

5 Consider the infinite series

$$S = \frac{9}{19} + \frac{99}{19^2} + \frac{999}{19^3} + \frac{9999}{19^4} + \cdots$$

In each term, the numerator is formed by writing the digit 9 repeatedly, and the denominator is a higher power of 19.

The series S converges.

What is the exact value of S ?

A $\frac{9}{19}$

B 1

C $\frac{20}{19}$

D $\frac{19}{18}$

E $\frac{10}{9}$

F $\frac{19}{9}$

- 6** In how many ways can 729 be written as a sum of two or more consecutive positive integers?
- A** 0
 - B** 5
 - C** 6
 - D** 7
 - E** 8

7 Let $P(x) = (2x^2 - 3x + 2)^3$.

What is the sum of the coefficients of the even powers of x in the expansion of $P(x)$? (Count the constant term as an even power of x .)

A -171

B 1

C 8

D 171

E 172

F 343

G 344

- 8 For integers n with $0 \leq n \leq 20$, consider the expression $\log_2 \left(\frac{n-9}{n-2} \right)$. For how many such integers n is this expression defined as a real number and has an integer value?

A 0

B 1

C 2

D 3

E 4

- 9** The curve C has equation $y = x^3 - 6x^2 + 4x$. The tangent to C at the point where $x = 1$ intersects C again at the point P . The tangent to C at P intersects C again at the point Q . Determine the x -coordinate of Q .

A -8

B -4

C -2

D 2

E 4

F 8

- 10** What is the shortest distance between the graphs of the following two equations in the xy -plane?

$$|x| + |y| = 6$$

$$|x - 1| + |y - 4| = 13$$

A 1

B $\sqrt{2}$

C 2

D $\sqrt{10}$

E $\sqrt{17}$

F 6

G $7\sqrt{2}$

11 The sum S_n is defined by

$$S_n = \sum_{r=1}^n \log_2 \left(1 + \frac{2r+3}{r(r+2)} \right)$$

Find the smallest integer n such that $S_n \geq 4 + \log_2 5$.

A 7

B 8

C 12

D 13

E 14

F 15

G 19

12 Consider the equation:

$$4 \sin x \cos x + 2\sqrt{2} \cos x = k(2 \sin x + \sqrt{2})$$

Find the sum of the squares of all possible real values for the constant k such that the equation has exactly three distinct solutions for x in the range $0 \leq x < 2\pi$.

A 3

B 4

C 6

D 8

E 10

F 12

13 Let $f(y) = y^2 - 2$.

Find the number of distinct real solutions x to the equation $f(f(f(|x|))) = 0$

A 2

B 4

C 6

D 8

E 12

F 16

- 14 Let a and b be positive real numbers. For every real x the following inequality holds:
 $a \leq \frac{4}{1+e^x} + \frac{1}{1+e^{-x}} + 1 \leq b$. What is the minimum possible value of $b - a$?
- A 1
 - B 2
 - C 3
 - D 4
 - E 5
 - F 21
 - G 23

- 15** For a real number x , let $[x]$ denote the largest integer less than or equal to x . Find the value of

$$\int_{-4}^4 (6x^2 + [x^3]) \, dx$$

- A** -8
- B** -4
- C** 124
- D** 252
- E** 256
- F** 260

Counts towards your Daily Integral streak.

16 Evaluate the sum

$$\frac{1}{\log_2(64!)} + \frac{1}{\log_3(64!)} + \frac{1}{\log_4(64!)} + \cdots + \frac{1}{\log_{64}(64!)}.$$

A 1

B 2

C 4

D 8

E 16

F 32

G 64

17 Let

$$S = \sum_{r=0}^{19} (-1)^r \binom{32}{r} \binom{32}{13+r}.$$

Evaluate S .

A 0

B $\binom{32}{9}$

C $\binom{32}{10}$

D $\binom{32}{13}$

E $\binom{32}{15}$

F $\binom{32}{16}$

- 18** Let θ be a real number. Define two infinite series:

$$S_1(\theta) = \sin^2 \theta + \sin^4 \theta + \sin^6 \theta + \dots$$

$$S_2(\theta) = \cos^2 \theta + \cos^4 \theta + \cos^6 \theta + \dots$$

Both series converge whenever $|\sin \theta| < 1$ and $|\cos \theta| < 1$.

Suppose that

$$S_1(\theta) = \frac{2}{3} \quad \text{and} \quad S_2(\theta) = \frac{3}{2}.$$

How many distinct real values of θ are there in the range $-2\pi \leq \theta \leq 2\pi$ which satisfy both equations?

A 2

B 4

C 6

D 8

E 10

19 Find the sum of all distinct real solutions to the following equation:

$$(x^2 - 7x + 11)^{x^2 - 2x - 2} = 1$$

A 7

B 9

C 11

D 13

E 14

F 16

- 20** Let the “bimode” of a data set be equal to the regular mode if there is a singular most common value, but equal to the mean of the modes if there is no unique modal value. What is the sum of all real values of x for which the mean, median and bimode (in any order) of the seven numbers

$$-2, 1, 1, 9, 9, 12, x$$

form an arithmetic sequence?

- A** -46
- B** 5
- C** 10
- D** 15
- E** 20
- F** 25
- G** 61
- H** 66

If stuck, a two-hour walk without your phone has been known to help.

Answer key

Question	Answer
1	C
2	A
3	C
4	E
5	D
6	C
7	E
8	D
9	C
10	B
11	E
12	F
13	D
14	C
15	D
16	A
17	A
18	D
19	D
20	F

20/20 with no mistakes? Did you just bang out a perfect run? What a god.

Anything else: don't worry too much. No one is really expecting 20 out of 20 on this one.